

Homology theory and Homotopy Invariance

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Abstract

Homology theory provides an algebraic invariant that assigns to every topological space a sequence of abelian groups, one for each non-negative dimension, in such a way that homeomorphic spaces have isomorphic homology groups. Unlike the fundamental group, which is a powerful but computationally difficult invariant that captures only low-dimensional information effectively, singular homology groups are comparatively tractable to compute even in higher dimensions and can distinguish spaces that the fundamental group cannot.

In this thesis we develop the theory of singular homology from first principles. We construct the singular chain complex associated to a topological space, verify that the boundary maps satisfy $\partial \circ \partial = 0$, and define the resulting homology groups $H_n(X)$. The central result of this thesis is the homotopy invariance theorem: if two spaces X and Y are homotopy equivalent, then $H_n(X) \cong H_n(Y)$ for every n . We give a complete proof of this theorem via the construction of an explicit chain homotopy. We then compute the homology groups of several standard spaces – the circle S^1 , the torus T , and the spheres S^n – using Δ -complex structures. We present two classical applications of these computations: invariance of dimension for Euclidean spaces, and the Brouwer fixed-point theorem. Finally, we develop relative homology, the long exact sequence of a pair, and excision, and use these tools to prove the equivalence of simplicial and singular homology and to establish the Mayer–Vietoris sequence.

Keywords: Singular homology, chain complex, homotopy invariance, Δ -complex, Brouwer fixed-point theorem, invariance of dimension, relative homology, excision, Mayer–Vietoris sequence.

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Chapter 1

Introduction and Motivation

In topology, one of the fundamental questions is to determine whether two given topological spaces X and Y are homeomorphic. Since directly constructing (or ruling out) a homeomorphism between two spaces is in general very difficult, topologists instead look for invariants: algebraic objects (groups, rings, numbers) associated to a space in such a way that homeomorphic — or even merely homotopy equivalent — spaces receive isomorphic (or equal) invariants. If two spaces have different invariants, they cannot be homeomorphic.

One of the earliest and most important such invariants is the fundamental group $\pi_1(X, x_0)$, which records the homotopy classes of loops based at a point $x_0 \in X$. The fundamental group is extremely effective for distinguishing low-dimensional spaces: for instance, it easily shows that S^1 is not simply connected while $\mathbb{R}^2$ is, and it underlies the classification of surfaces. However, the fundamental group has serious limitations:

- It is often very difficult, or even algorithmically undecidable in general, to determine whether two given finitely presented groups are isomorphic.
- For $n \geq 2$, the higher homotopy groups $\pi_n(X)$ exist and are of interest, but they are notoriously difficult to compute — even for simple spaces such as spheres, the groups $\pi_k(S^n)$ for $k > n$ remain an active area of research.
- The fundamental group by itself fails to distinguish higher-dimensional spaces effectively; for example, it cannot be used directly to show that $\mathbb{R}^3 \not\cong \mathbb{R}^4$, since both are simply connected.

This motivates the search for invariants that are easier to compute while still being sensitive to the topology of the space in all dimensions. Homology groups $H_n(X)$, $n \geq 0$, are exactly such invariants. They are always abelian (unlike π_1 , which need not be), they are defined in every dimension simultaneously, and — crucially — they can be computed algorithmically via linear algebra once a suitable combinatorial (simplicial or Δ -complex) structure on the space is known.

The goal of this thesis is to:

1. Construct singular homology groups $H_n(X)$ for an arbitrary topological space X (Chapter 2, Chapter 3).
2. Prove that homology is a homotopy invariant: homotopy equivalent spaces have isomorphic homology groups (Chapter 4).

3. Compute the homology groups of some standard spaces: S^1 , the torus, and S^n (Chapter 5).
4. Derive two classical consequences: topological invariance of dimension, and the Brouwer fixed-point theorem (Chapter 6).
5. Develop relative homology, the long exact sequence of a pair, and excision, and use them to prove the equivalence of simplicial and singular homology and to establish the Mayer–Vietoris sequence (Chapters 7–9).

Chapter 2

Chain Complexes and Singular Homology

2.1 Singular simplices and singular chains

Let Δ^n denote the standard n -simplex,

$$\Delta^n = \left\{ (t_0, \dots, t_n) \in \mathbb{R}^{n+1} : t_i \geq 0, \sum_{i=0}^n t_i = 1 \right\},$$

with vertices $v_0, \dots, v_n$, where v_i is the point with $t_i = 1$ and all other coordinates 0.

Definition 2.1. Let X be a topological space. A *singular n -simplex* in X is a continuous map

$$\sigma : \Delta^n \rightarrow X.$$

Note that a singular simplex need not be injective, nor need its image resemble a geometric simplex at all; it is simply a continuous parametrized “probe” of X by the standard simplex.

Definition 2.2. The *singular n -chain group* $C_n(X)$ is the free abelian group generated by the set of all singular n -simplices in X . Its elements, called *singular n -chains*, are finite formal sums

$$c = \sum_j n_j \sigma_j, \quad n_j \in \mathbb{Z}, \quad \sigma_j : \Delta^n \rightarrow X.$$

2.2 The boundary map

For a sequence of vertices $v_0, \dots, v_n$ (not necessarily distinct as points of X , but formally indexed), we write $[v_0, \dots, v_n]$ for the affine singular simplex sending the i -th vertex of Δ^n to v_i . The notation $[v_0, \dots, \hat{v}_i, \dots, v_n]$ denotes the $(n-1)$ -simplex obtained by omitting the vertex v_i ; this is the i -th face of the simplex.

Definition 2.3. The *boundary homomorphism* $\partial_n : C_n(X) \rightarrow C_{n-1}(X)$ is defined on a singular n -simplex $\sigma : \Delta^n \rightarrow X$ by

$$\partial_n(\sigma) = \sum_{i=0}^n (-1)^i \sigma|_{[v_0, \dots, \hat{v}_i, \dots, v_n]},$$

and extended to all of $C_n(X)$ by linearity.

Here $\sigma|_{[v_0, \dots, \hat{v}_i, \dots, v_n]}$ denotes the restriction of σ to the i -th face of Δ^n , viewed as a singular $(n-1)$ -simplex via the canonical identification of that face with Δ^{n-1} .

Proposition 2.4. *The composition*

$$C_n(X) \xrightarrow{\partial_n} C_{n-1}(X) \xrightarrow{\partial_{n-1}} C_{n-2}(X)$$

is the zero map, i.e., $\partial_{n-1} \circ \partial_n = 0$.

Proof. It suffices to check this on a generator σ . Expanding,

$$\partial_n(\sigma) = \sum_i (-1)^i \sigma|_{[v_0, \dots, \hat{v}_i, \dots, v_n]},$$

and applying ∂_{n-1} term by term, splitting according to whether the next omitted index j satisfies $j < i$ or $j > i$,

$$\partial_{n-1}\partial_n(\sigma) = \sum_{j < i} (-1)^i (-1)^j \sigma|_{[v_0, \dots, \hat{v}_j, \dots, \hat{v}_i, \dots, v_n]} + \sum_{j > i} (-1)^i (-1)^{j-1} \sigma|_{[v_0, \dots, \hat{v}_i, \dots, \hat{v}_j, \dots, v_n]}.$$

In the second sum, relabel the pair (i, j) as (j, i) (swapping the roles of the two omitted indices, which is legitimate since each term records an unordered pair of omitted vertices together with a sign). Under this relabeling, the second sum becomes

$$\sum_{j < i} (-1)^j (-1)^{i-1} \sigma|_{[v_0, \dots, \hat{v}_j, \dots, \hat{v}_i, \dots, v_n]} = - \sum_{j < i} (-1)^i (-1)^j \sigma|_{[v_0, \dots, \hat{v}_j, \dots, \hat{v}_i, \dots, v_n]},$$

which is exactly the negative of the first sum. Hence the two sums cancel and $\partial_{n-1}\partial_n(\sigma) = 0$. $\square$

2.3 Chain complexes

Definition 2.5. A *chain complex* of abelian groups $C_\bullet$ consists of a sequence of abelian groups C_n , $n \geq 0$, together with homomorphisms $\partial_n : C_n \rightarrow C_{n-1}$ for $n \geq 1$, satisfying $\partial_{n-1} \circ \partial_n = 0$ for all $n \geq 2$. Such a complex is depicted as

$$\dots \xrightarrow{\partial_{n+1}} C_{n+1} \xrightarrow{\partial_n} C_n \xrightarrow{\partial_{n-1}} \dots \xrightarrow{\partial_1} C_0.$$

By Proposition 2.4, the groups $C_n(X)$ together with the maps ∂_n form a chain complex, called the *singular chain complex* of X , denoted $C_\bullet(X)$.

Chapter 3

Homology Groups

3.1 Cycles, boundaries, and homology

Let $C_\bullet = (C_n, \partial_n)$ be a chain complex.

Definition 3.1. The group of n -cycles is

$$Z_n = Z_n(C) := \ker(\partial_n : C_n \rightarrow C_{n-1}) \subseteq C_n,$$

and the group of n -boundaries is

$$B_n = B_n(C) := \operatorname{im}(\partial_{n+1} : C_{n+1} \rightarrow C_n) \subseteq C_n.$$

By convention $Z_0 = C_0$, i.e., every 0-chain is regarded as a 0-cycle.

Since $\partial_n \circ \partial_{n+1} = 0$, every boundary is a cycle:

$$B_n \subseteq Z_n \quad \text{for all } n \geq 0.$$

Definition 3.2. The n -th homology group of the chain complex C is the quotient group

$$H_n(C) := Z_n(C)/B_n(C).$$

Elements of $H_n(C)$ are cosets $z_n + B_n(C)$ with $z_n \in Z_n(C)$, called *homology classes*; two cycles representing the same homology class are said to be *homologous*.

Definition 3.3. For a topological space X , the n -th singular homology group of X is

$$H_n(X) := H_n(C_\bullet(X)) = Z_n(X)/B_n(X),$$

where $Z_n(X)$ and $B_n(X)$ are the cycle and boundary groups of the singular chain complex $C_\bullet(X)$.

Intuitively, an n -cycle is a singular n -chain with empty boundary; it represents a closed “surface” of dimension n mapped into X . It is a boundary if it is itself the boundary of some $(n+1)$ -chain, i.e., if it “bounds” a higher-dimensional piece inside X . The homology group measures the extent to which cycles fail to be boundaries — equivalently, it detects n -dimensional “holes” in X .

Chapter 4

Homotopy Invariance of Homology

The main theorem of this thesis is that singular homology is invariant under homotopy equivalence, not merely homeomorphism. This is a much stronger and more useful statement, since homotopy equivalence is a coarser (easier to establish) relation than homeomorphism.

Theorem 4.1. *If X and Y are homotopy equivalent topological spaces, then $H_n(X) \cong H_n(Y)$ for every $n \geq 0$.*

The proof proceeds in three steps: first we show that a continuous map induces a chain map and hence a homomorphism on homology (Proposition 4.2); second, we show that homotopic maps induce the same homomorphism on homology (Proposition 4.3); third, we combine these facts using the definition of homotopy equivalence.

4.1 Induced homomorphisms

Proposition 4.2. *A continuous map $f : X \rightarrow Y$ induces a chain map*

$$f_{\#} : C_n(X) \rightarrow C_n(Y), \quad n \geq 0,$$

defined on a generator $\sigma : \Delta^n \rightarrow X$ by $f_{\#}(\sigma) = f \circ \sigma$ and extended linearly. Moreover $f_{\#}$ commutes with the boundary map, $\partial \circ f_{\#} = f_{\#} \circ \partial$, so $f_{\#}$ carries cycles to cycles and boundaries to boundaries, and hence induces a well-defined homomorphism

$$f_* : H_n(X) \rightarrow H_n(Y), \quad f_*(\alpha + B_n^X(C)) = f_{\#}(\alpha) + B_n^Y(C).$$

Proof. That $f_{\#}$ is a chain map (i.e., commutes with ∂) follows directly from the definitions, since restricting $f \circ \sigma$ to a face of Δ^n is the same as applying f to the restriction of σ to that face. Since $f_{\#}$ commutes with ∂ , it sends $\ker \partial_n$ into $\ker \partial_n$ and $\operatorname{im} \partial_{n+1}$ into $\operatorname{im} \partial_{n+1}$; hence it descends to a well-defined map on the quotient $H_n(X) = Z_n(X)/B_n(X) \rightarrow Z_n(Y)/B_n(Y) = H_n(Y)$. $\square$

Functoriality is immediate: $(\operatorname{Id}_X)_* = \operatorname{Id}_{H_n(X)}$, and for continuous maps $f : X \rightarrow Y$, $g : Y \rightarrow Z$,

$$(g \circ f)_* = g_* \circ f_*.$$

4.2 Chain homotopy and invariance under homotopic maps

Proposition 4.3. *If $f, g : X \rightarrow Y$ are homotopic continuous maps, $f \simeq g$, then the induced maps on homology coincide:*

$$f_* = g_* : H_n(X) \rightarrow H_n(Y) \quad \text{for every } n.$$

Proof. Let $F : X \times I \rightarrow Y$ be a homotopy with $F(x, 0) = f(x)$ and $F(x, 1) = g(x)$ for all $x \in X$. Write $\Delta^n \times I$ as the union of prisms with vertices $v_0, \dots, v_n$ on $\Delta^n \times \{0\}$ and $w_0, \dots, w_n$ on $\Delta^n \times \{1\}$.

We construct a chain homotopy $P : C_n(X) \rightarrow C_{n+1}(Y)$: for a singular simplex $\sigma : \Delta^n \rightarrow X$, define

$$P(\sigma) = \sum_{i=0}^n (-1)^i F \circ (\sigma, \text{Id})|_{[v_0, \dots, v_i, w_i, \dots, w_n]} \in C_{n+1}(Y),$$

and extend linearly to all chains: $P(\sum_j c_j \sigma_j) = \sum_j c_j P(\sigma_j)$. Here $F \circ (\sigma, \text{Id})$ denotes the composite

$$\Delta^n \times I \xrightarrow{(\sigma, \text{Id})} X \times I \xrightarrow{F} Y.$$

This subdivides the prism $\Delta^n \times I$ into $(n+1)$ singular $(n+1)$ -simplices, with alternating signs, in the standard way.

Claim. $\partial P + P\partial = g_\# - f_\#$ as maps $C_n(X) \rightarrow C_n(Y)$.

We verify this by direct computation in the case $n = 1$; the general case follows the same pattern with more elaborate bookkeeping of the prism boundary. Let $\sigma : [v_0, v_1] \rightarrow X$. Then

$$P(\sigma) = F \circ (\sigma, \text{Id})|_{[v_0, v_1, w_1]} - F \circ (\sigma, \text{Id})|_{[v_0, w_0, w_1]}.$$

Taking the boundary of each 2-simplex,

$$\begin{aligned} \partial P(\sigma) &= F \circ (\sigma, \text{Id})|_{[v_1, w_1]} - F \circ (\sigma, \text{Id})|_{[v_0, w_1]} + F \circ (\sigma, \text{Id})|_{[v_0, v_1]} \\ &\quad - F \circ (\sigma, \text{Id})|_{[w_0, w_1]} + F \circ (\sigma, \text{Id})|_{[v_0, w_1]} - F \circ (\sigma, \text{Id})|_{[v_0, w_0]}. \end{aligned}$$

The two terms $\pm F \circ (\sigma, \text{Id})|_{[v_0, w_1]}$ cancel. The term $F \circ (\sigma, \text{Id})|_{[v_0, v_1]}$ equals $f_\#(\sigma)$ evaluated appropriately, and $F \circ (\sigma, \text{Id})|_{[w_0, w_1]}$ equals $g_\#(\sigma)$, so after identifying the remaining edge terms with $P(\partial\sigma)$ (since $\partial\sigma = v_1 - v_0$, and the prism over a point v_i collapses to the segment $[v_i, w_i]$), we obtain

$$\partial P(\sigma) = g_\#(\sigma) - f_\#(\sigma) - P(\partial\sigma),$$

i.e., $\partial P(\sigma) + P(\partial\sigma) = g_\#(\sigma) - f_\#(\sigma)$, as claimed. The identity for general n is proved by the analogous computation applied to the full prism decomposition.

Now take any homology class $\alpha + B_n^X(C) \in H_n(X)$, represented by a cycle α , so $\partial\alpha = 0$. Then

$$(g_\# - f_\#)(\alpha) = P(\partial\alpha) + \partial(P(\alpha)) = \partial(P(\alpha)),$$

which lies in $B_n^Y(C)$. Hence

$$g_*(\alpha + B_n^X(C)) - f_*(\alpha + B_n^X(C)) = (g_\#(\alpha) - f_\#(\alpha)) + B_n^Y(C) = 0 + B_n^Y(C),$$

so $f_* = g_*$. □

4.3 Proof of the main theorem

Proof of Theorem 4.1. Since X and Y are homotopy equivalent, there exist continuous maps $\varphi : X \rightarrow Y$ and $\rho : Y \rightarrow X$ such that

$$\varphi \circ \rho \simeq \text{Id}_Y, \quad \rho \circ \varphi \simeq \text{Id}_X.$$

By functoriality of the induced map (Proposition 4.2),

$$(\varphi \circ \rho)_* = \varphi_* \circ \rho_*, \quad (\rho \circ \varphi)_* = \rho_* \circ \varphi_*.$$

By Proposition 4.3, homotopic maps induce equal homomorphisms on homology, so

$$(\varphi \circ \rho)_* = (\text{Id}_Y)_* = \text{Id}_{H_n(Y)}, \quad (\rho \circ \varphi)_* = (\text{Id}_X)_* = \text{Id}_{H_n(X)}.$$

Combining these,

$$\varphi_* \circ \rho_* = \text{Id}_{H_n(Y)}, \quad \rho_* \circ \varphi_* = \text{Id}_{H_n(X)},$$

so $\varphi_* : H_n(X) \rightarrow H_n(Y)$ is an isomorphism with inverse ρ_* . This holds for every $n \geq 0$, completing the proof. $\square$

Remark 4.4. In particular, homeomorphic spaces are homotopy equivalent, so Theorem 4.1 implies that H_n is a topological invariant: it is exactly the tool we set out to construct in Chapter 1.

Chapter 5

Homology Groups of Some Standard Spaces

In this chapter we compute the homology groups of several spaces using Δ -complex structures, which give a finite combinatorial chain complex $\Delta_\bullet(X)$ computing $H_n(X)$.

5.1 The circle S^1

Give S^1 the Δ -complex structure with a single vertex v and a single edge e (a loop). Then $\Delta_0(S^1) \cong \mathbb{Z}$ and $\Delta_1(S^1) \cong \mathbb{Z}$, both generated by a single basis element, and $\Delta_n(S^1) = 0$ for $n \geq 2$, since there are no simplices of dimension ≥ 2 in this structure. The boundary map ∂_1 sends $e \mapsto v - v = 0$, so $\partial_1 = 0$. Hence $Z_1 = \Delta_1(S^1) \cong \mathbb{Z}$ and $B_1 = 0$, while $Z_0 = \Delta_0(S^1) \cong \mathbb{Z}$ and $B_0 = \text{im } \partial_1 = 0$. Therefore

$$H_n(S^1) \cong \begin{cases} \mathbb{Z} & n = 0, 1 \\ 0 & n \geq 2. \end{cases}$$

5.2 The torus $T = S^1 \times S^1$

Give the torus T the Δ -complex structure obtained from a square with edges identified in the usual pattern: one vertex v , three edges a, b, c , and two 2-simplices U (upper) and L (lower), where c is the diagonal of the square.

Since all three edges begin and end at v , we have $\partial_1(a) = \partial_1(b) = \partial_1(c) = v - v = 0$, so $\partial_1 = 0$. The 2-simplices have boundaries

$$\partial_2(U) = c - b - a, \quad \partial_2(L) = b + a - c,$$

so that the chain complex is

$$0 \rightarrow \Delta_2(T) \xrightarrow{\partial_2} \Delta_1(T) \xrightarrow{\partial_1=0} \Delta_0(T) \rightarrow 0.$$

$H_0(T)$: Since $\partial_1 = 0$, we have $\ker \partial_0 = \Delta_0(T) \cong \mathbb{Z}$ and $\text{im } \partial_1 = 0$, so

$$H_0(T) = \Delta_0(T)/0 \cong \mathbb{Z}.$$

$H_1(T)$: Since $\partial_1 = 0$, $Z_1 = \Delta_1(T) \cong \mathbb{Z}\langle a, b, c \rangle \cong \mathbb{Z}^3$. The boundary group $B_1 = \text{im } \partial_2$ is generated by $\partial_2(U) = c - b - a$ and $\partial_2(L) = b + a - c = -(c - b - a)$; thus B_1 is the infinite cyclic subgroup generated by $c - a - b$. Quotienting,

$$H_1(T) = Z_1/B_1 \cong \mathbb{Z}\langle a, b, c \mid a + b = c \rangle \cong \mathbb{Z} \times \mathbb{Z},$$

since imposing the single relation $c = a + b$ eliminates c as an independent generator, leaving a and b free.

$H_2(T)$: There are no 3-simplices, so $H_2(T) = \ker \partial_2$. If $pU + qL \in \ker \partial_2$, then

$$\partial_2(pU + qL) = (p - q)(a + b - c) = 0$$

forces $p = q$ (since $a + b - c \neq 0$ in the free group $\Delta_1(T)$). Hence $\ker \partial_2$ is infinite cyclic, generated by $U - L$ (using the convention $\partial_2 L = -(a + b - c)$; equivalently $U + L$ depending on sign convention), so

$$H_2(T) \cong \mathbb{Z}.$$

$H_n(T) = 0$ for $n \geq 3$, since $\Delta_n(T) = 0$ in this range. Summarizing,

$$H_n(T) \cong \begin{cases} \mathbb{Z} \times \mathbb{Z} & n = 1 \\ \mathbb{Z} & n = 0, 2 \\ 0 & n \geq 3. \end{cases}$$

5.3 Spheres S^n

Give S^2 the Δ -complex structure consisting of two copies of Δ^2 , labelled U and L , with their boundaries identified via the identity map. Label the shared edges a, b, c and the shared vertices u, v, w . Then

$$\partial_2(U) = a + b + c, \quad \partial_2(L) = -(a + b + c),$$

$$\partial_1(a) = v - u, \quad \partial_1(b) = w - v, \quad \partial_1(c) = u - w.$$

$H_0(S^2)$: $\ker \partial_0 = \Delta_0(S^2)$; since u, v, w are all connected by edges, $\text{im } \partial_1$ identifies $u = v = w$ in the quotient, giving

$$H_0(S^2) \cong \mathbb{Z}.$$

$H_1(S^2)$: $\ker \partial_1$ consists of chains $\lambda a + \mu b + \nu c$ with $\lambda(v - u) + \mu(w - v) + \nu(u - w) = 0$; since u, v, w are linearly independent generators of Δ_0 , this forces $\lambda = \mu = \nu$, so $\ker \partial_1$ is generated by $a + b + c$. But $\text{im } \partial_2$ is also generated by $a + b + c$ (up to sign). Hence

$$H_1(S^2) \cong 0.$$

$H_2(S^2)$: There are no 3-simplices, so $H_2(S^2) = \ker \partial_2$. If $pU + qL \in \ker \partial_2$, then $\partial_2(pU + qL) = (p - q)(a + b + c) = 0$ forces $p = q$, so $\ker \partial_2$ is infinite cyclic, generated by $U + L$, giving

$$H_2(S^2) \cong \mathbb{Z}.$$

Thus

$$H_n(S^2) \cong \begin{cases} 0 & n = 1 \\ \mathbb{Z} & n = 0, 2 \\ 0 & n \geq 3. \end{cases}$$

The same construction generalizes to S^n for any $n \geq 1$: using two copies of Δ^n glued along their boundary, an identical argument (there being no $(n+1)$ -simplices, so $H_n(S^n) = \ker \partial_n$, which is infinite cyclic generated by $U + L$ since $\partial_n(pU + qL) = (p - q)(a + b + c) = 0$ forces $p = q$) yields

$$H_q(S^n) \cong \begin{cases} 0 & q \neq 0, n \\ \mathbb{Z} & q = 0, n, \end{cases}$$

for $n > 0$.

For $n = 0$, the space S^0 consists of two points, so trivially

$$H_0(S^0) \cong \mathbb{Z} \times \mathbb{Z}, \quad H_q(S^0) \cong 0 \text{ for } q \neq 0.$$

Chapter 6

Applications

6.1 Invariance of dimension

Corollary 6.1. *If $m \neq n$, then S^m and S^n are not of the same homotopy type.*

Proof. By the computations of Chapter 5, $H_n(S^m)$ is isomorphic to $H_n(S^n) \cong \mathbb{Z}$ only when $m = n$; if $m \neq n$ then $H_m(S^m) \cong \mathbb{Z}$ while $H_m(S^n) \cong 0$ (assuming, without loss of generality, $m < n$). By Theorem 4.1, homotopy equivalent spaces have isomorphic homology groups in every dimension, so $S^m \not\cong S^n$. $\square$

Corollary 6.2 (Invariance of dimension). *Two Euclidean spaces E^m and E^n are homeomorphic if and only if $m = n$.*

Proof. If $m = n$ the identity map is a homeomorphism. Conversely, suppose $h : E^m \rightarrow E^n$ is a homeomorphism. Removing the point $h(0)$ from E^n and 0 from E^m , h restricts to a homeomorphism $E^m \setminus \{0\} \cong E^n \setminus \{h(0)\}$. Since $E^m \setminus \{0\}$ deformation retracts onto S^{m-1} (via radial projection) and similarly $E^n \setminus \{h(0)\}$ deformation retracts onto S^{n-1} , we get a chain of homotopy equivalences

$$S^{m-1} \simeq E^m \setminus \{0\} \cong E^n \setminus \{h(0)\} \simeq S^{n-1}.$$

By Corollary 6.1, this forces $m - 1 = n - 1$, i.e., $m = n$. $\square$

6.2 The Brouwer fixed-point theorem

Let B^n denote the closed unit ball in $\mathbb{R}^n$.

Theorem 6.3 (Brouwer Fixed-Point Theorem). *Every continuous map $f : B^n \rightarrow B^n$ has at least one fixed point.*

Proof. Suppose, for contradiction, that $f : B^n \rightarrow B^n$ is continuous and has no fixed point, so that $x \neq f(x)$ for every $x \in B^n$.

Case $n = 1$. Here $B^1 = [0, 1]$. Define $g : [0, 1] \rightarrow \{-1, 1\}$ by

$$g(t) = \frac{t - f(t)}{|t - f(t)|},$$

which is well-defined and continuous since $t \neq f(t)$ for all t . But $[0, 1]$ is connected, while its continuous image under g would have to be connected; $\{-1, 1\}$ is disconnected. This is a contradiction.

Case $n = 2$. Since $f(x) \neq x$ for all $x \in B^2$, one can define a retraction $r : B^2 \rightarrow S^1$ by sending x to the point where the ray from $f(x)$ through x meets S^1 ; this r is continuous and satisfies $r(x) = x$ for $x \in S^1$. Fixing a basepoint $p = (1, 0) \in S^1$ and letting $i : S^1 \hookrightarrow B^2$ denote inclusion, we have $r \circ i = \text{Id}_{S^1}$, so on fundamental groups

$$\pi_1(S^1, p) \xrightarrow{i_*} \pi_1(B^2, p) \xrightarrow{r_*} \pi_1(S^1, p)$$

satisfies $r_* \circ i_* = \text{Id}$, forcing r_* to be surjective. But $\pi_1(B^2, p) = 0$ (since B^2 is convex, hence contractible) while $\pi_1(S^1, p) \cong \mathbb{Z}$, and there is no surjection from the trivial group onto $\mathbb{Z}$. Contradiction.

Case $n \geq 2$ (homological argument, generalizing to all n). As in the case $n = 2$, the assumption that f has no fixed point produces a retraction $r : B^n \rightarrow S^{n-1}$: for each $x \in B^n$, let $r(x)$ be the point where the ray starting at $f(x)$ and passing through x exits B^n ; continuity of r follows from continuity of f and the fact that $x \neq f(x)$. If $i : S^{n-1} \hookrightarrow B^n$ is the inclusion, then $r \circ i = \text{Id}_{S^{n-1}}$, i.e., S^{n-1} is a retract of B^n .

By Proposition 6.4 below, this forces $i_* : H_{n-1}(S^{n-1}) \rightarrow H_{n-1}(B^n)$ to be injective. But by the computations of Chapter 5, $H_{n-1}(S^{n-1}) \cong \mathbb{Z}$, while $H_{n-1}(B^n) = 0$ since B^n is convex and hence contractible (a contractible space is homotopy equivalent to a point, and $H_k(\text{point}) = 0$ for $k \geq 1$ by Theorem 4.1). An injective homomorphism from $\mathbb{Z}$ into the trivial group is impossible. This contradiction completes the proof. $\square$

Proposition 6.4. *If $i : A \hookrightarrow X$ is the inclusion of a retract A of X (i.e., there is a continuous $r : X \rightarrow A$ with $r \circ i = \text{Id}_A$), then*

$$i_* : H_k(A) \rightarrow H_k(X)$$

is injective (a monomorphism) for every k .

Proof. Applying the functor $H_k(-)$ to the identity $r \circ i = \text{Id}_A$ and using functoriality of induced maps (Proposition 4.2),

$$r_* \circ i_* = (r \circ i)_* = (\text{Id}_A)_* = \text{Id}_{H_k(A)}.$$

Since $\text{Id}_{H_k(A)}$ is injective, and a composition $r_* \circ i_*$ can only be injective if i_* itself is injective, it follows that i_* is injective. $\square$

Chapter 7

Relative Homology and the Long Exact Sequence of a Pair

7.1 Relative chains and relative homology

Definition 7.1. Let $A \subseteq X$ be a subspace. Since $C_n(A) \subseteq C_n(X)$ is a subgroup (the free abelian group on singular n -simplices landing entirely in A) and the boundary map ∂_n restricts to $C_n(A)$, we may form the quotient chain complex

$$C_n(X, A) := C_n(X) / C_n(A),$$

with induced boundary map $\bar{\partial}_n : C_n(X, A) \rightarrow C_{n-1}(X, A)$, well-defined because $\partial_n(C_n(A)) \subseteq C_{n-1}(A)$. The homology groups of this complex,

$$H_n(X, A) := \ker \bar{\partial}_n / \operatorname{im} \bar{\partial}_{n+1},$$

are the *relative homology groups* of the pair (X, A) .

Concretely, a relative cycle is a chain $\alpha \in C_n(X)$ with $\partial\alpha \in C_n(A)$ (not necessarily $\partial\alpha = 0$), and it is a relative boundary if $\alpha = \partial\beta + \gamma$ for some $\beta \in C_{n+1}(X)$, $\gamma \in C_n(A)$.

7.2 The algebraic zig-zag lemma

The engine behind both results in this chapter and the next is a purely algebraic fact.

Proposition 7.2 (Zig-zag / Snake Lemma for chain complexes). *Let*

$$0 \rightarrow A_\bullet \xrightarrow{i} B_\bullet \xrightarrow{p} C_\bullet \rightarrow 0$$

be a short exact sequence of chain complexes (i.e. $0 \rightarrow A_n \rightarrow B_n \rightarrow C_n \rightarrow 0$ is exact for every n , and i, p commute with the boundary maps). Then there is a long exact sequence of homology groups

$$\cdots \rightarrow H_n(A) \xrightarrow{i_*} H_n(B) \xrightarrow{p_*} H_n(C) \xrightarrow{\delta} H_{n-1}(A) \xrightarrow{i_*} H_{n-1}(B) \rightarrow \cdots$$

Proof. The maps i_*, p_* are the ones induced functorially by chain maps (Proposition 4.2 applies verbatim to any chain map, not just those induced by continuous maps). It remains to construct δ and check exactness at the three spots.

Construction of δ . Let $[c] \in H_n(C)$ be represented by a cycle $c \in C_n$, $\partial c = 0$. Since $p_n : B_n \rightarrow C_n$ is surjective, choose $b \in B_n$ with $p(b) = c$. Then $p(\partial b) = \partial p(b) = \partial c = 0$, so $\partial b \in \ker p_{n-1} = \text{im } i_{n-1}$; choose $a \in A_{n-1}$ with $i(a) = \partial b$. Since i is injective and $i(\partial a) = \partial i(a) = \partial \partial b = 0$, we get $\partial a = 0$, so a is a cycle. Define $\delta[c] := [a] \in H_{n-1}(A)$.

Well-definedness. If b' is another lift of c , then $b - b' \in \ker p_n = \text{im } i_n$, say $b - b' = i(a_0)$. Then the corresponding $a' \in A_{n-1}$ satisfies $i(a - a') = \partial(b - b') = i(\partial a_0)$, and since i is injective, $a - a' = \partial a_0 \in \text{im } \partial$; hence $[a] = [a']$ in $H_{n-1}(A)$. If c is replaced by a homologous cycle $c + \partial c_1$ with $c_1 \in C_{n+1}$, lift c_1 to $b_1 \in B_{n+1}$; then $b + \partial b_1$ is a valid lift of $c + \partial c_1$, and it produces the same class a in A_{n-1} modulo boundaries, by the same computation. So δ is well-defined on homology classes.

Exactness at $H_n(B)$. $p_* i_* = (pi)_* = 0_* = 0$ since $pi = 0$, so $\text{im } i_* \subseteq \ker p_*$. Conversely, suppose $[b] \in \ker p_*$, i.e. $p(b) = \partial c'$ for some $c' \in C_{n+1}$. Lift c' to $b' \in B_{n+1}$; then $p(b - \partial b') = p(b) - \partial p(b') = \partial c' - \partial c' = 0$, so $b - \partial b' = i(a)$ for some $a \in A_n$. Since $i(\partial a) = \partial i(a) = \partial(b - \partial b') = \partial b = 0$ (as b is a cycle) and i is injective, a is a cycle, and $i_*[a] = [i(a)] = [b - \partial b'] = [b]$. Hence $\ker p_* \subseteq \text{im } i_*$.

Exactness at $H_n(C)$. If $[c] = p_*[b]$ for a cycle b , then in the construction of $\delta[c]$ we may take the lift of c to be b itself, giving $\partial b = 0$ and hence $a = 0$ (since $i(a) = \partial b = 0$ and i injective), so $\delta[c] = 0$. Thus $\text{im } p_* \subseteq \ker \delta$. Conversely, if $\delta[c] = 0$, then in the construction $a = \partial a_0$ for some $a_0 \in A_n$; then $b - i(a_0)$ satisfies $\partial(b - i(a_0)) = \partial b - i(\partial a_0) = i(a) - i(a) = 0$, so $b - i(a_0)$ is a cycle in B_n with $p_*[b - i(a_0)] = [p(b)] - 0 = [c]$. Thus $\ker \delta \subseteq \text{im } p_*$.

Exactness at $H_{n-1}(A)$. By construction, $\delta[c] = [a]$ where $i(a) = \partial b$ for a lift b of a cycle c ; hence $i_* \delta[c] = [i(a)] = [\partial b] = 0$ in $H_{n-1}(B)$, giving $\text{im } \delta \subseteq \ker i_*$. Conversely if $i_*[a] = 0$, i.e. $i(a) = \partial b$ for some $b \in B_n$, set $c := p(b) \in C_n$; then $\partial c = p(\partial b) = p(i(a)) = 0$, so c is a cycle, and by construction $\delta[c] = [a]$. Thus $\ker i_* \subseteq \text{im } \delta$. $\square$

7.3 The long exact sequence of a pair

Since $0 \rightarrow C_n(A) \rightarrow C_n(X) \rightarrow C_n(X, A) \rightarrow 0$ is a short exact sequence of chain complexes for every subspace $A \subseteq X$ (with the inclusion and quotient maps commuting with ∂ by construction), Proposition 7.2 applies immediately:

Theorem 7.3. *For any pair (X, A) there is a long exact sequence*

$$\cdots \rightarrow H_n(A) \xrightarrow{i_*} H_n(X) \xrightarrow{j_*} H_n(X, A) \xrightarrow{\delta} H_{n-1}(A) \rightarrow \cdots \rightarrow H_0(X, A) \rightarrow 0,$$

where i is inclusion and j is the quotient map on chains.

This is the tool that will let us relate the homology of a skeletal filtration piece by piece, and (with excision) will drive the Mayer–Vietoris sequence.

7.4 Excision

The final ingredient is the fact that homology does not see subspaces cut out of the “interior” of a pair — this is what lets us compare small pieces of a space to the whole.

Theorem 7.4 (Excision). *Given subspaces $Z \subseteq A \subseteq X$ with $\overline{Z} \subseteq \text{int}(A)$, the inclusion $(X \setminus Z, A \setminus Z) \hookrightarrow (X, A)$ induces isomorphisms*

$$H_n(X \setminus Z, A \setminus Z) \xrightarrow{\cong} H_n(X, A) \quad \text{for all } n.$$

Proof sketch. The proof rests on the *subdivision theorem*: if $X = \text{int}(A) \cup \text{int}(B)$ for subspaces $A, B \subseteq X$, then the inclusion of the subcomplex $C_n^{A,B}(X) \subseteq C_n(X)$ generated by singular simplices whose image lies entirely in A or entirely in B is a chain homotopy equivalence with $C_n(X)$ — every singular simplex is chain-homotopic to a sum of “small” simplices lying in A or B , obtained by iterated barycentric subdivision (each subdivision shrinks simplices, and since $\{\text{int } A, \text{int } B\}$ is an open cover of the compact set Δ^n ’s image, finitely many subdivisions suffice by a Lebesgue-number argument). This produces an explicit chain homotopy T with $\partial T + T\partial = \text{id} - S$ where S is the iterated subdivision operator, landing S inside $C_n^{A,B}(X)$. One then applies excision with $Z = X \setminus (A \cup B)$: taking $B = X \setminus Z$ (so $A \cup B = X$ by hypothesis $\bar{Z} \subseteq \text{int } A$), the induced map $C_\bullet(A) + C_\bullet(B) \rightarrow C_\bullet(X)$ is a quasi-isomorphism, and passing to the quotient by $C_\bullet(A)$ gives the isomorphism on relative homology. We omit the full bookkeeping of the subdivision operator, which is lengthy but routine (see Hatcher, *Algebraic Topology*, §2.1). $\square$

Chapter 8

Equivalence of Simplicial and Singular Homology

8.1 Simplicial chains for Δ -complexes

Recall from Chapter 5 that a Δ -complex structure on X gives, for each n , a set of n -simplices $\{\sigma_\alpha : \Delta^n \rightarrow X\}$ glued along faces. Let X^n denote the n -skeleton (union of images of simplices of dimension $\leq n$).

Definition 8.1. The *simplicial chain group* $\Delta_n(X)$ is the free abelian group on the (finitely or infinitely many) n -simplices of the Δ -complex structure, with boundary map defined exactly as in Definition 2.3, restricted to these simplices. The resulting homology $H_n^\Delta(X) := \ker \partial_n^\Delta / \text{im } \partial_{n+1}^\Delta$ is the *simplicial homology* of X . (This is precisely the complex used in the computations of Chapter 5.)

There is an evident chain map $\Delta_\bullet(X) \hookrightarrow C_\bullet(X)$, since every simplicial n -simplex σ_α is in particular a singular n -simplex $\Delta^n \rightarrow X$, and this inclusion commutes with ∂ .

8.2 The comparison theorem

Theorem 8.2. For a Δ -complex X , the inclusion $\Delta_\bullet(X) \hookrightarrow C_\bullet(X)$ induces an isomorphism

$$H_n^\Delta(X) \xrightarrow{\cong} H_n(X) \quad \text{for every } n.$$

Proof. We work with the relative groups $H_n^\Delta(X^k, X^{k-1})$ and $H_n(X^k, X^{k-1})$ and compare skeleton by skeleton, using the long exact sequence of Theorem 7.3.

Step 1: The relative groups agree in each skeleton. By construction, $\Delta_n(X^k, X^{k-1}) = \Delta_n(X^k) / \Delta_n(X^{k-1})$ is zero unless $n = k$, in which case it is free abelian on the k -simplices of X , with zero boundary map (since ∂^Δ of a k -simplex lands in $\Delta_{k-1}(X^{k-1})$, which becomes 0 in the quotient). Hence

$$H_n^\Delta(X^k, X^{k-1}) \cong \begin{cases} \bigoplus_\alpha \mathbb{Z} \langle \sigma_\alpha \rangle & n = k \\ 0 & n \neq k. \end{cases}$$

On the singular side, one shows $H_n(X^k, X^{k-1})$ has the same description: X^k / X^{k-1} is (up to homeomorphism, via the characteristic maps of the Δ -complex) a wedge of k -spheres, one for each k -simplex, obtained by collapsing the boundary of each k -simplex to

a point; using excision (Theorem 7.4) to excise the complement of a neighborhood of each attached cell, together with the computation $H_n(S^k)$ from Chapter 5, one obtains the identical description $H_n(X^k, X^{k-1}) \cong \bigoplus_{\alpha} \mathbb{Z}$ concentrated in degree $n = k$, one summand per k -simplex. Under these identifications the inclusion of chain complexes induces the identity map on these groups.

Step 2: Passing from skeleta to X by induction and a colimit argument.

Consider the commutative ladder of long exact sequences (Theorem 7.3) for the pair (X^k, X^{k-1}) , comparing the simplicial and singular versions via the natural transformation induced by inclusion of chain complexes:

$$\begin{array}{ccccccccc} H_n^{\Delta}(X^{k-1}) & \rightarrow & H_n^{\Delta}(X^k) & \rightarrow & H_n^{\Delta}(X^k, X^{k-1}) & \rightarrow & H_{n-1}^{\Delta}(X^{k-1}) & \rightarrow & H_{n-1}^{\Delta}(X^k) \\ \downarrow & & \downarrow & & \downarrow \cong & & \downarrow & & \downarrow \\ H_n(X^{k-1}) & \rightarrow & H_n(X^k) & \rightarrow & H_n(X^k, X^{k-1}) & \rightarrow & H_{n-1}(X^{k-1}) & \rightarrow & H_{n-1}(X^k) \end{array}$$

By induction on k (base case $X^{-1} = \emptyset$, where both sides vanish trivially), assume the leftmost and rightmost pairs of vertical maps are isomorphisms for all n . Since the middle vertical map is an isomorphism by Step 1, the *five lemma* (an elementary diagram chase, exactly analogous to the argument used in the proof of Proposition 6.4, applied twice) shows the map $H_n^{\Delta}(X^k) \rightarrow H_n(X^k)$ is also an isomorphism for every n . This completes the induction.

Step 3: Passing to the limit. If X has infinitely many simplices (infinite-dimensional or infinitely many cells in some dimension), every singular chain and every simplicial chain involves only finitely many simplices, hence factors through some finite skeleton X^k ; likewise every homology relation (a chain being a boundary) is witnessed inside some finite-dimensional subcomplex. Formally, $C_{\bullet}(X) = \varinjlim_k C_{\bullet}(X^k)$ and $\Delta_{\bullet}(X) = \varinjlim_k \Delta_{\bullet}(X^k)$ as directed colimits, and homology commutes with directed colimits of chain complexes (since $\ker$ and coker do). Combined with Step 2, this gives $H_n^{\Delta}(X) \cong H_n(X)$ for all n . $\square$

Corollary 8.3. *The computations in Chapter 5 (of $H_n(S^1)$, $H_n(T)$, $H_n(S^n)$ via Δ -complexes) are computations of the genuine singular homology groups, not merely of an auxiliary invariant.*

Chapter 9

The Mayer–Vietoris Sequence

9.1 Statement

Theorem 9.1 (Mayer–Vietoris). *Let $X = \text{int}(A) \cup \text{int}(B)$ for subspaces $A, B \subseteq X$. There is a long exact sequence*

$$\cdots \rightarrow H_n(A \cap B) \xrightarrow{\psi} H_n(A) \oplus H_n(B) \xrightarrow{\varphi} H_n(X) \xrightarrow{\delta} H_{n-1}(A \cap B) \rightarrow \cdots \rightarrow H_0(X) \rightarrow 0,$$

where $\psi(x) = (i_*x, -j_*x)$ and $\varphi(a, b) = k_*a + l_*b$, for the four inclusions $i : A \cap B \hookrightarrow A$, $j : A \cap B \hookrightarrow B$, $k : A \hookrightarrow X$, $l : B \hookrightarrow X$.

9.2 Construction and proof

Proof. Recall from the excision proof sketch (Theorem 7.4) that under the hypothesis $X = \text{int}(A) \cup \text{int}(B)$, the subcomplex

$$C_n^{A,B}(X) := C_n(A) + C_n(B) \subseteq C_n(X)$$

(the subgroup generated by chains lying entirely in A together with chains lying entirely in B) includes as a chain homotopy equivalence into $C_\bullet(X)$; in particular, $H_n(C_\bullet^{A,B}(X)) \cong H_n(X)$ for every n , via the map induced by inclusion.

Consider the sequence of chain groups

$$0 \rightarrow C_n(A \cap B) \xrightarrow{\Psi} C_n(A) \oplus C_n(B) \xrightarrow{\Phi} C_n^{A,B}(X) \rightarrow 0,$$

where $\Psi(x) = (i_\#x, -j_\#x)$ and $\Phi(a, b) = k_\#a + l_\#b$ (using the notation for induced chain maps from Proposition 4.2).

Exactness. Φ is surjective by definition of $C_n^{A,B}(X)$ as a sum. Ψ is injective since a chain in $A \cap B$ that is zero as a chain in A is certainly zero. For exactness in the middle: if $\Phi(a, b) = k_\#a + l_\#b = 0$ in $C_n(X)$, then writing each generator explicitly, $a = -b$ as chains in $C_n(X)$; but a is (by definition) a sum of singular simplices lying in A , and $-b$ a sum of singular simplices lying in B , and their equality as formal sums in $C_n(X)$ forces every simplex appearing to lie in both A and B (matching basis elements of the free abelian group $C_n(X)$ term by term), i.e. $a = -b = x$ for some $x \in C_n(A \cap B)$, so $(a, b) = \Psi(x)$. This is a short exact sequence of chain complexes (the maps visibly commute with ∂).

Applying the zig-zag lemma (Proposition 7.2) yields the long exact sequence

$$\cdots \rightarrow H_n(A \cap B) \xrightarrow{\Psi_*} H_n(A) \oplus H_n(B) \xrightarrow{\Phi_*} H_n(C_{\bullet}^{A,B}(X)) \xrightarrow{\delta} H_{n-1}(A \cap B) \rightarrow \cdots$$

Composing the isomorphism $H_n(C_{\bullet}^{A,B}(X)) \cong H_n(X)$ from the excision/subdivision theorem with Φ_* identifies Φ_* with the map φ of the statement, and Ψ_* with ψ , giving exactly the asserted sequence. $\square$

9.3 Sample application: recomputing $H_*(S^n)$

As a check consistent with Chapter 5, write $S^n = A \cup B$ where A, B are the (slightly enlarged, open) upper and lower hemispheres, each contractible, with $A \cap B \simeq S^{n-1}$ (an equatorial band). Since $H_k(A) = H_k(B) = 0$ for $k \geq 1$ (contractible spaces), the Mayer–Vietoris sequence collapses for $k \geq 2$ to isomorphisms

$$H_k(S^n) \cong H_{k-1}(S^{n-1}),$$

and a short exact sequence at the bottom handles $k = 0, 1$. Inducting on n from the base case $H_*(S^0)$ (computed directly in Chapter 5) reproduces $H_k(S^n) \cong \mathbb{Z}$ for $k = 0, n$ and 0 otherwise — matching Chapter 5’s Δ -complex computation, now obtained without any explicit simplicial structure.

Conclusion

We have constructed the singular chain complex of a topological space and the associated homology groups, proved that these groups are invariant under homotopy equivalence, and used explicit Δ -complex structures to compute the homology groups of S^1 , the torus, and S^n for all n . These computations were then leveraged to prove two classical results in topology: the topological invariance of dimension for Euclidean spaces, and the Brouwer fixed-point theorem. We then developed relative homology and the long exact sequence of a pair, together with the excision theorem, and used these tools to prove that simplicial and singular homology agree on Δ -complexes, and to construct the Mayer–Vietoris sequence, illustrated by a recomputation of the homology of spheres. These examples illustrate the central theme of algebraic topology — translating topological problems into algebraic ones that can be resolved by direct computation.

Natural directions for further study include cellular homology and its equivalence with singular homology, cohomology and cup products, and the relationship between singular homology and other generalized homology theories, all of which extend the computational reach of the methods introduced here.

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